

Yara Ismail Alsiamy

Artificial Intelligence Graduate | Machine Learning Engineer | Business Analyst

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Artificial Intelligence graduate with First Class Honors from Umm Al-Qura University and a member of the university's first graduating cohort in Artificial Intelligence. Experienced in Machine Learning, Computer Vision, NLP, Digital Twin Systems, IoT, and Business Analysis through the national-scale Nusuk Card Platform. Designed and deployed award-winning AI solutions, including a real-time Digital Twin system for smart greenhouses. Passionate about building intelligent systems that create measurable real-world impact.

PROFESSIONAL EXPERIENCE

Business Analyst Intern - Nusuk Card (Hajj 1446), Experts Vision Consulting | EVC

Mar 2025 – Jul 2025

- Gathering requirements, preparing BRD and SRS, and developing use cases to support system development.
- Contributed to system documentation (user manuals) and participated in system testing and validation.
- Executed data entry and validation processes for workers on the Nusuk Card platform, ensuring high data accuracy and completeness.
- Monitored and supported Nusuk card operational workflows, including tracking processes to ensure timely execution.
- Coordinated with internal teams and stakeholders to resolve operational issues and maintain smooth workflow continuity.

EDUCATION

Bachelor of Science in Artificial Intelligence, Umm Al-Qura University

2022 – 2026

First Class Honors | GPA: 3.84/4

Member of Umm Al-Qura University's First Graduating Cohort in Artificial Intelligence

PROJECTS

Graduation Project | Warif – AI Digital Twin for Smart Farms

Oct 2025 – Jun 2026

Project Demo: [📺](#)

Award-winning AI- and IoT-powered Digital Twin system for smart greenhouse management that continuously synchronizes real-time sensor data to generate AI-driven recommendations, autonomous decisions, and direct control of irrigation and climate devices, optimizing resource utilization and supporting sustainable agriculture.

- Successfully deployed and validated the system in a real cucumber greenhouse.
- Integrated IoT sensors and smart devices for real-time monitoring and autonomous environmental control.
- Achieved 29.81% water reduction, 35% energy reduction, and 41% improvement in irrigation efficiency.
- Designed, developed, and managed a cloud-hosted PostgreSQL database for real-time data synchronization.
- Designed and developed responsive frontend interfaces using React.js.
- Awarded 1st Place – Best Artificial Intelligence Graduation Project (Injaz 2026).

Arabic Sign Language Recognition System

Dec 2025

Designed and developed a real-time web-based Arabic Sign Language recognition system using YOLOv8, supporting live webcam streaming, image upload, and image capture.

- Designed the AI system architecture and deployment workflow.
- Implemented modular architecture with version control and real-time monitoring.

Traffic Vehicle Detection using YOLOv8

Dec 2025

Fine-tuned a YOLOv8 model for real-time traffic vehicle detection using transfer learning and hyperparameter optimization. Improved detection performance through StepLR scheduling, Label Smoothing, and data augmentation, achieving mAP@50 of 0.867 and mAP@50–95 of 0.737.

Generative Image Modeling using VAEs & GANs

May 2026

Implemented and evaluated baseline and modified VAE and GAN models for image generation using the Fashion-MNIST dataset. Compared model performance, training stability, and image quality, including the effect of label smoothing on GAN training.

Email Intent Classification using Transformer Encoder

May 2026

Developed a Transformer encoder for multi-class email intent classification using the Enron Email Dataset. Implemented text preprocessing, tokenization, attention masking, and evaluated the model using Accuracy and Macro F1-score with an ablation study on attention heads.

Human Segmentation from Images

Oct 2024

Developed a MATLAB-based human segmentation application using edge detection, thresholding, and color-based segmentation techniques. Built an interactive GUI to compare segmentation methods and visualize results.

AWARDS

1st Place – Best Artificial Intelligence Graduation Project (Injaz 2026), College of Computing, Umm Al-Qura University

Jun 2026

Awarded for developing "Warif", an AI- and IoT-powered Digital Twin system for smart greenhouse management, enabling real-time monitoring, autonomous decision-making, and sustainable resource optimization.

1st place - Health and Security Hackathon for Hajj, Ministry of interior

Jun 2024

Designing infrastructure models that support the health and safety of the security man and the pilgrim

SKILLS

Technical Skills

- **Artificial Intelligence:** Machine Learning, Deep Learning, Computer Vision, Natural Language Processing (NLP), Generative AI, Digital Twin Systems, Internet of Things (IoT)
- **AI & Backend Technologies:** TensorFlow, PyTorch, Transformers, FastAPI, MQTT, REST APIs
- **Business Analysis:** Requirements Gathering, BRD, SRS, Use Cases, System Testing
- **Programming:** Python, Java, JavaScript, React.js, C, HTML/CSS, Shell Scripting
- **Data & Databases:** PostgreSQL, SQL, Database Design, Data Analysis, Data Cleaning, Data Visualization
- **Development Tools:** Git, GitHub, VS Code, Jupyter Notebook, Google Colab, MATLAB, Figma, Draw.io, Microsoft 365
- **Mathematics:** Linear Algebra, Probability & Statistics, Calculus for Machine Learning

LANGUAGES

Arabic - English